



Problem Statements for CRAFATHON'26

Problem Statement 1: Medication Non-Adherence

Theme: HealthTech

Objective:

To develop a monitoring and alert system that tracks patient medication adherence, detects missed or incorrect doses, and provides timely reminders and reports to prevent silent treatment failure.

Problem Description:

Medication adherence is critical for successful treatment outcomes, especially for chronic diseases. However, patients frequently miss doses, stop medication early, or take incorrect dosages due to forgetfulness, misunderstanding, or lack of supervision. Healthcare providers often have no real-time visibility into patient adherence. This project proposes a smart system that monitors medication intake behavior using digital records, sensor data, or user interaction logs, identifies non-compliance patterns, and generates alerts or reports for patients and caregivers.

Core Features:

- Medication schedule management
- Detection of missed or delayed doses
- Adherence tracking and compliance scoring
- Automated reminders and alerts
- Non-adherence pattern analysis
- Reporting dashboard for caregivers or doctors
- Scalable for chronic and long-term treatments

Datasets Required:

- **Electronic Health Records (EHR) – Synthetic Dataset (MIMIC-III)**
<https://physionet.org/content/mimiciii/1.4/>
- **Medication Event Monitoring System (MEMS) – Sample Data**
<https://physionet.org/content/mimiciii-demo/>

Problem Statement 2: Smart Appointment Scheduling

Theme: HealthTech

Objective:

To develop an intelligent appointment management system that optimizes scheduling, reduces patient waiting time, and improves overall operational efficiency in healthcare facilities.

Problem Description:

Traditional appointment systems operate on fixed schedules and do not account for real-time delays, variable consultation times, emergency cases, or no-shows. As a result, patients experience long waiting periods while doctors face uneven workloads. This project proposes a data-driven appointment management system that analyzes historical appointment data and real-time inputs to predict consultation durations, dynamically adjust schedules, and provide accurate waiting time estimates to patients.

Core Features:

- Intelligent appointment scheduling and rescheduling
- Prediction of consultation duration
- Real-time waiting time estimation
- No-show and delay handling
- Queue optimization and load balancing
- Doctor availability and utilization tracking
- Patient notification and alert system

Datasets Required:

- **Hospital Appointment Dataset (Kaggle)**
<https://www.kaggle.com/datasets/joniarroba/noshowappointments>